Data Appendix

Predicting Wildfires from Time Series Weather Data

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1. Dataset Overview

This appendix documents the dataset used to predict wildfire occurrence in California using daily weather measurements. The dataset spans **January 1, 1984 to January 10, 2025** and contains **14,976 observations** after removing 12 rows with missing values.

Unit of Observation: Each row represents a single calendar day in California. It records weather conditions measured that day (precipitation, high/low temperature, wind speed) along with engineered features derived from those measurements and a binary indicator of whether a wildfire started on that date.

Data Sources:
• National Oceanic and Atmospheric Administration (NOAA) Climate Data Online — provides the daily weather measurements.
• California Department of Forestry and Fire Protection (CAL FIRE) — provides wildfire event records.
• Dataset assembled and published on Zenodo as an integrated daily record.

2. Variable Descriptions

2.1 DATE

Type	date
Description	Date of the daily observation.
Range / Values	1984-01-01 to 2025-01-10
Notes	No missing values after cleaning.

2.2 PRECIPITATION

Type	continuous (inches)
Description	Total precipitation recorded on that day.

Range / Values	Min: 0.000, Max: 4.530, Mean: 0.032
Notes	12 rows with missing weather data were dropped during cleaning.

2.3 MAX_TEMP

Type	continuous (degrees F)
Description	Maximum temperature recorded during the day.
Range / Values	Min: 50.0, Max: 106.0, Mean: 70.5
Notes	Positive trend observed over the dataset period.

2.4 MIN_TEMP

Type	continuous (degrees F)
Description	Minimum temperature recorded during the day.
Range / Values	Min: 33.0, Max: 77.0, Mean: 56.5
Notes	Rising faster than MAX_TEMP; nights are warming more than daytime highs.

2.5 AVG_WIND_SPEED

Type	continuous (mph)
Description	Average wind speed across the day.
Range / Values	Min: 1.79, Max: 26.17, Mean: 7.44
Notes	Highly correlated with WIND_TEMP_RATIO.

2.6 FIRE_START_DAY

Type	binary (0/1)
Description	Indicates whether a wildfire started on this day. 0 = no wildfire; 1 = at least one wildfire started.
Range / Values	0 (No Fire): 10005 days, 1 (Fire): 4971 days
Notes	Target variable. Imbalanced: ~33% positive class.

2.7 YEAR

Type	integer
Description	Calendar year of the observation.
Range / Values	1984 to 2025
Notes	Used as a feature to capture long-term temporal trends.

2.8 MONTH

Type	integer (1-12)
Description	Calendar month of the observation.
Range / Values	1 (January) to 12 (December)
Notes	Used as a feature to capture seasonality.

2.9 TEMP_RANGE

Type	continuous (degrees F)
Description	Difference between MAX_TEMP and MIN_TEMP.
Range / Values	Min: 2.0, Max: 41.0, Mean: 14.0
Notes	Correlated with MAX_TEMP; warmer days tend to have larger temperature swings.

2.10 WIND_TEMP_RATIO

Type	continuous (mph/F)
Description	Ratio of AVG_WIND_SPEED to MAX_TEMP.
Range / Values	Min: 0.0236, Max: 0.4591, Mean: 0.1070
Notes	Engineered feature; highly collinear with AVG_WIND_SPEED.

2.11 LAGGED_PRECIPITATION

Type	continuous (inches)
Description	Cumulative precipitation over the preceding 7 days.
Range / Values	Min: 0.000, Max: 8.180, Mean: 0.226
Notes	Captures drought or wet-period conditions leading up to each day.

2.12 LAGGED_AVG_WIND_SPEED

Type	continuous (mph)
Description	Average wind speed over the preceding 7 days.
Range / Values	Min: 3.23, Max: 13.93, Mean: 7.43
Notes	Rolling wind context for each observation.

2.13 SEASON_ENC

Type	ordinal integer (0-3)

Description	Season encoded as an integer: 0=Winter, 1=Spring, 2=Summer, 3=Fall.
Range / Values	0, 1, 2, or 3
Notes	Derived from original SEASON string column which was dropped.

3. Summary Statistics

The table below presents descriptive statistics for all numeric variables in the analyzed dataset (N = 14,976).

Statistic	PRECIP	MAX_T	MIN_T	AVG_WND	T_RANGE	WND/T	LAG_PRCP	LAG_WND	FIRE
count	14976.0000	14976.0000	14976.0000	14976.0000	14976.0000	14976.0000	14976.0000	14976.0000	14976.0000
mean	0.0323	70.5409	56.4997	7.4351	14.0413	0.1070	0.2264	7.4340	0.3319
std	0.1796	7.2620	6.7661	2.1300	5.9961	0.0356	0.6489	1.3850	0.4709
min	0.0000	50.0000	33.0000	1.7900	2.0000	0.0236	0.0000	3.2271	0.0000
25%	0.0000	65.0000	51.0000	6.0400	10.0000	0.0852	0.0000	6.5186	0.0000
50%	0.0000	70.0000	57.0000	7.1600	12.0000	0.1022	0.0000	7.4786	0.0000
75%	0.0000	75.0000	62.0000	8.5000	17.0000	0.1205	0.0600	8.2786	1.0000
max	4.5300	106.0000	77.0000	26.1700	41.0000	0.4591	8.1800	13.9329	1.0000

4. Exploratory Data Analysis Figures

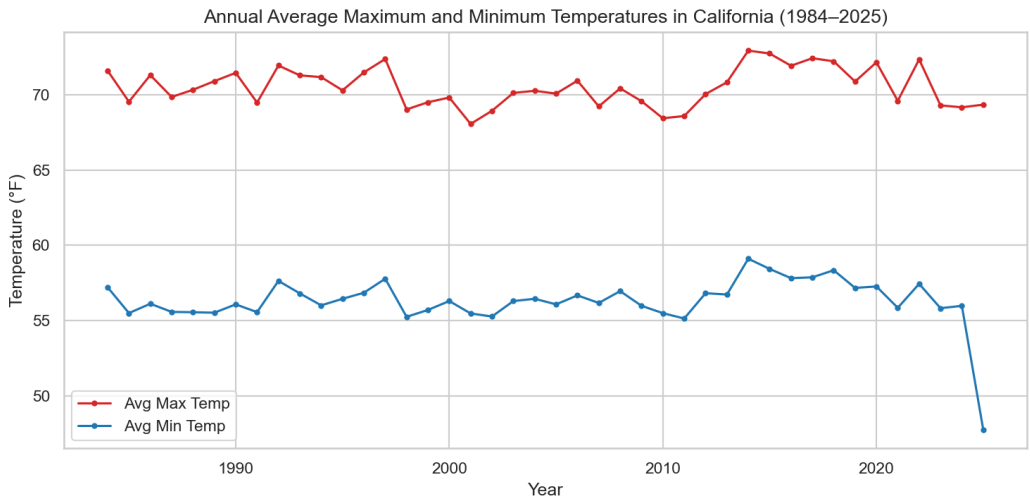

Figure 1: Annual average maximum and minimum temperatures (1984-2025). Both show a slight upward trend, with minimum temperatures rising more steeply than maximums.

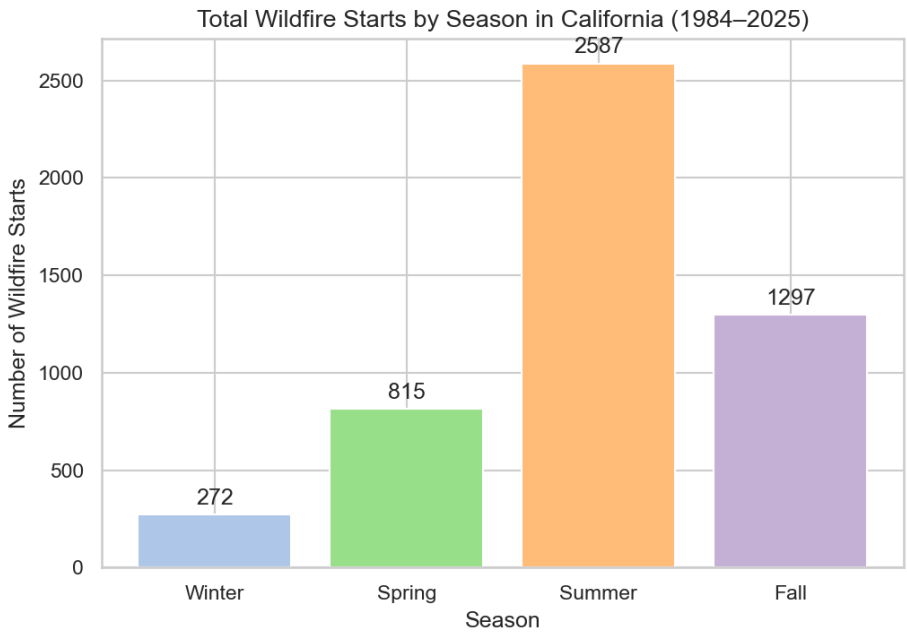

Figure 2: Total wildfire starts by season across the full dataset. Summer and Fall account for the majority of wildfire activity.

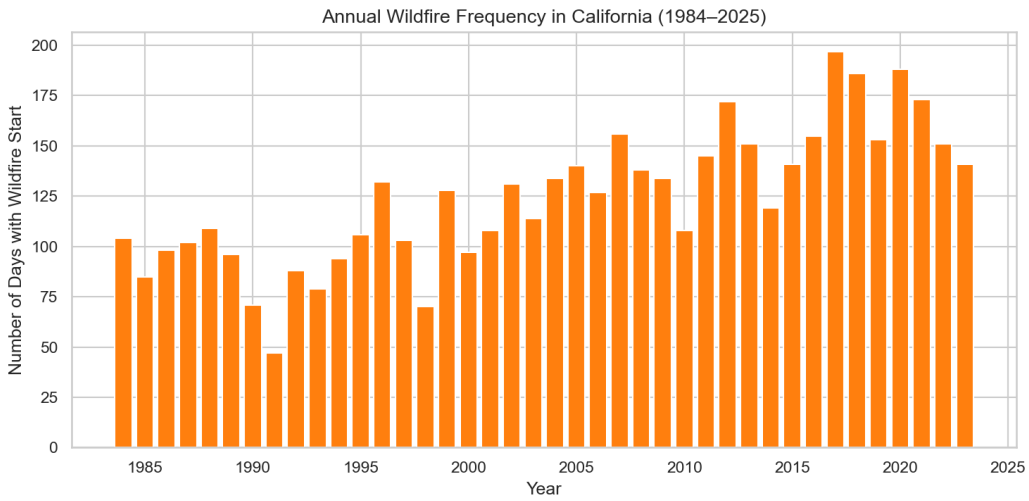

Figure 3: Annual wildfire frequency. Wildfire starts reached a peak of 197 in 2017. The partial year 2025 explains the sharp drop at the end.

Figure 4: Class distribution of the target variable FIRE_START_DAY. The dataset is moderately imbalanced at approximately 33% positive class.

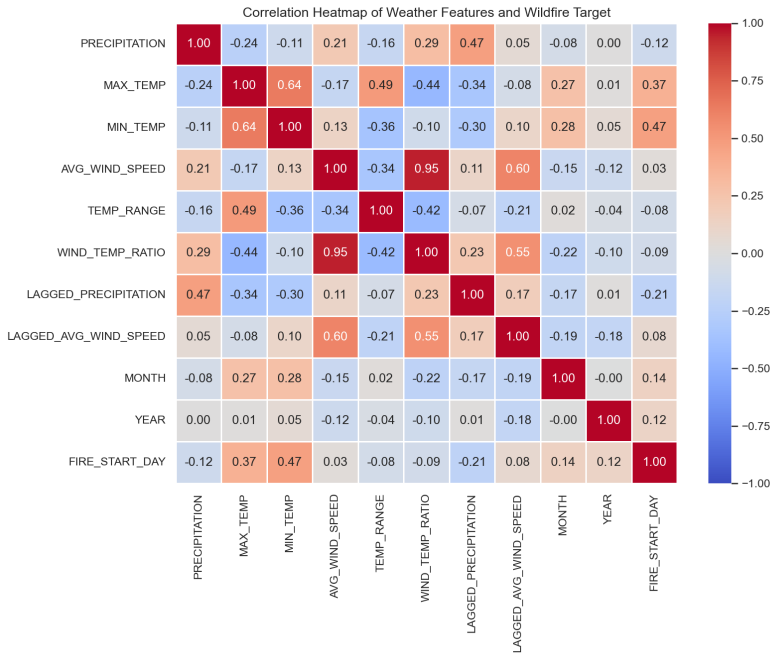

Figure 5: Correlation heatmap of numeric features. Notable correlations include AVG_WIND_SPEED and WIND_TEMP_RATIO, and MAX_TEMP with TEMP_RANGE.